

GROUP PROJECT OUTLINE

PRIMARY OUTCOME	What is the primary outcome? How are you <i>specifically</i> defining this outcome?
PRIMARY EXSPORE(S)	What is associated with your primary outcome?
Health Disparity	Based on individual characteristics (gender, age, LGBTQ+, SES, income, education, neighborhood, etc.), what is the health disparity? What is the magnitude and the direction of the health disparity?
Public Health Significance: Big Picture Findings based on 7-10 peer review research articles	Why do we care? How has the significance changed over time?
Public Health Significance: Local or specific to rural communities Findings based on 7-10 peer review research articles (different from above)	
Feasible solutions based on peer review research	Hypothesize potential solutions to your public health problem. What are successful interventions that have worked in other populations that could be implemented in our community? What are the challenges to implementing some of these potential solutions?

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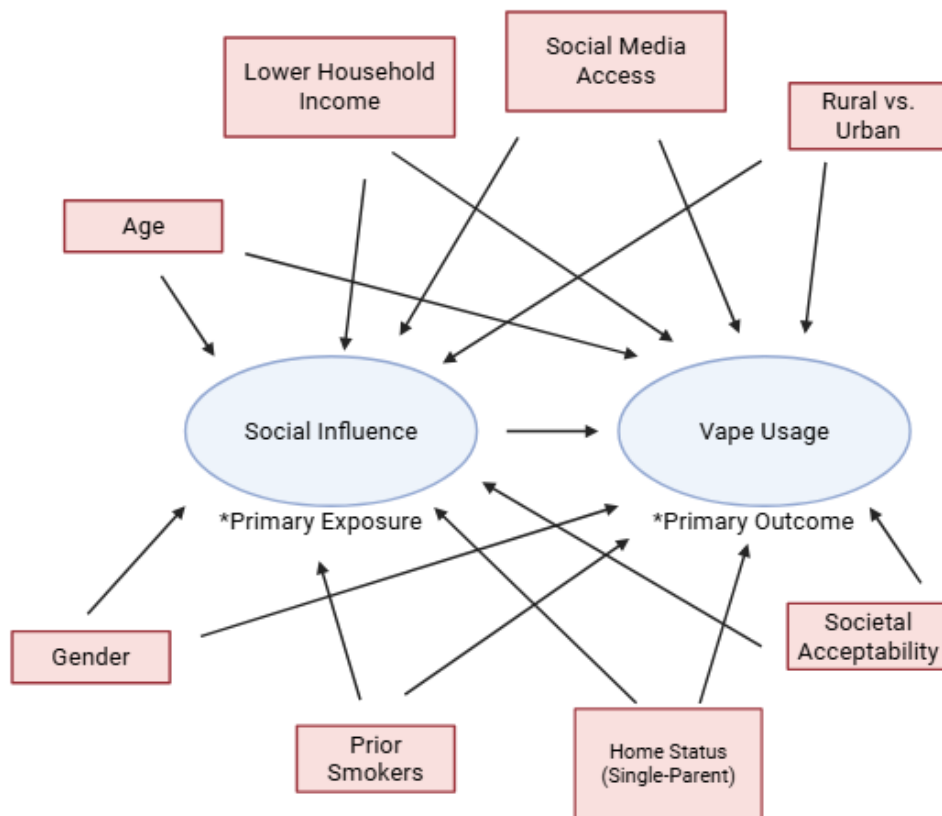
GROUP PROJECT OUTLINE: SUBSTANCE USE

PRIMARY OUTCOME	What is the primary outcome? Vape Usage How are you <i>specifically</i> defining this outcome? Individuals who vape in Total Service Area
PRIMARY EXSPORE(S)	What is associated with your primary outcome? Social Influence - Exposure to peer and familial figures actively vaping leading to encouragement of experimentation
Health Disparity	Based on individual characteristics (gender, age, LGBTQ+, SES, income, education, neighborhood, etc.), what is the health disparity? Age → Minors in TSA who are vaping What is the magnitude and the direction of the health disparity? Minors are highly susceptible
Public Health Significance: Big Picture Findings based on 7-10 peer review research articles	Why do we care? Minors/young adults are more vulnerable to developing addictions and addictive properties and are more vulnerable with regard to health consequences because of their underdeveloped brains. Because of this vulnerability, it could lead to more serious health issues later on in life. -about vaping in general -vaping in rural and in ky -vaping in youth -research problem is youth in rural communities How has the significance changed over time? Who is Vaping Vaping Confounders Vape Marketing Primary Outcome Gateway Drug Research Vape marketing and policy - Discussion section: “The proportion of youth who reported any exposure to marketing for vaping products was greater among youth living in countries with fewer marketing restrictions” National Youth Tobacco Survey Implications of the “Vaping Epidemic” in Youth
Public Health Significance: Local or specific to rural communities	E-cigarette use in Kentucky - Compares rates of vaping in the US with rates in kentucky and then looks specifically at Appalachian Kentucky - Gives evidence that vaping rates in kentucky are significantly higher than the vaping rates of the US as a

Findings based on 7-10 peer review research articles (different from above)	whole, and within that, young adults have the highest rates <u>Changes in smoking prevalence and opinions on policy among college students</u> - Looks at the prevalence of smoking among college students and the trends over a 4-year period - Talks about their reasons for smoking and opinions on policy change and how that has affected the use - Talks about a decline in cigarette usage but an increase in vape usage <u>Youth mental health and nicotine vape use: The moderating role of rural-urban/suburban school environments</u> - Discusses a connection to mental health - “findings showed that the relationship between nicotine vape use and mental health was stronger for youth attending high school in rural areas” <u>Engaging Rural High School Youth in E-cigarette Prevention and Advocacy</u> - Specifically looks at appalachian area - Looks at rural high school students - Gives insight into potential interventions <u>Rural-urban differences e-cigarette ever use, the perception of harm, and e-cigarette information seeking behaviors among U.S. adults in a nationally representative study</u> - Is specific to adults but discusses concept of trust in rural communities vs urban (trust in leaders and trust in tobacco companies) - Could be a confounder - Could also help when thinking of intervention <u>Rethinking Rural</u> - Gives insight into how public health advocates could strengthen their knowledge on E-cigarettes to eventually put an end to their usage among youth in rural areas <u>Addressing Rural Adolescent E-cigarette Use</u> - Addresses the health risks of nicotine and smoking - Talk about programs and resources that are being developed to address this issue
Feasible solutions based on peer review research	Hypothesize potential solutions to your public health problem. - Education intervention in grade and middle schools - Education intervention for parents (signs of what to look for, ways to intervene, tools to talk to their children) What are successful interventions that have worked in other populations that could be implemented in our community? - <u>One study</u> noted some success with digital based interventions - One study noted that policies and regulations have been proven effective (taxation) - One study also noted that public health initiatives and “aggressive education campaigns” have also been successful in traditional cigarette use and might be

	effective for vapes; they just need additional research. What are the challenges to implementing some of these potential solutions? - One of the main challenges has been the regulation of internet sales. Some smaller companies don't ask for age and identity confirmation, making it hard to limit the sale. Because of this, minors have been able to resell to others. - On a broader scale, it is hard to implement laws and regulations because of the long process through legislation.
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GROUP PROJECT OUTLINE: FOOD INSECURITY

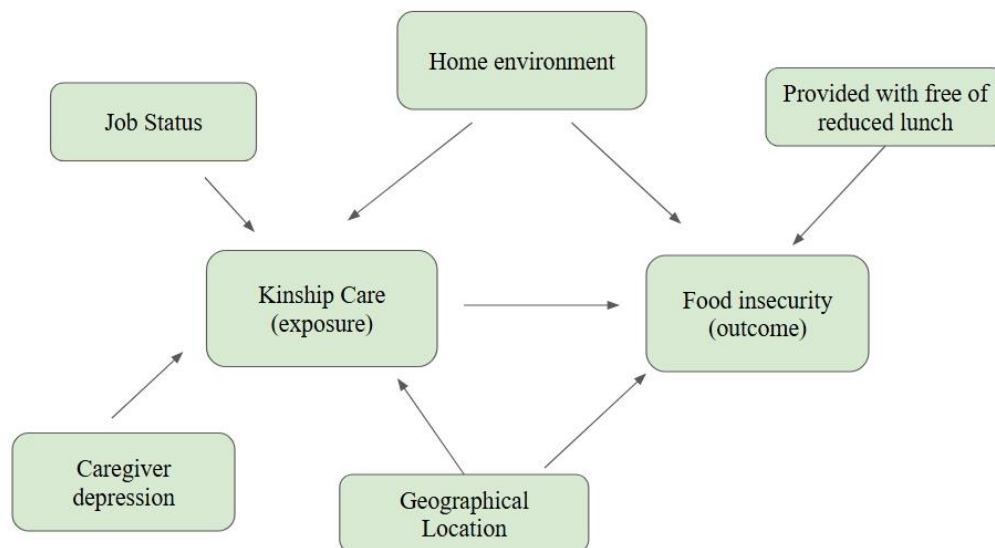
PRIMARY OUTCOME	What is the primary outcome? Food insecurity among school aged children in Kinship Care How are you <i>specifically</i> defining this outcome? School aged children (between 5-18) raised by grandparents or other relatives that report they worry about where their next meal will come from, specifically focusing on after school hours, weekends, and breaks from school.
PRIMARY EXPOSURE(S)	What is associated with your primary outcome? 1. Being raised by grandparents or other relatives (Kinship Care)
Health Disparity	Based on individual characteristics (gender, age, LGBTQ+, SES, income, education, neighborhood, etc.), what is the health disparity? Home environment, where children are raised, may influence access to healthy and affordable foods. Or generally care. Income, families and caregivers with lower and unstable income. Geographical location / Neighborhood, access to grocery stores and healthy and affordable foods for feeding and raising young children. Job status, if the caregiver is able to work or unable, can result in struggle to afford food or properly care for young children. Caregiver Depression, the result of financial burdens and other founders Children already provided with free or reduced lunch National immigration policies What is the magnitude and the direction of the health disparity? The magnitude is severe, and the issue is time sensitive. The direction of this health disparity will potentially be moving in the wrong direction with the removal of governmental resources in the form of Supplemental Nutrition Assistance Program (SNAP), and the strain on the Kentucky Transitional Assistance Program (KTAP) program.
Public Health	Why do we care?

Significance: Big Picture Findings based on 7-10 peer review research articles	We care because this current pressing issue could potentially worsen with current events, making people who are food secure more vulnerable to contracting health issues. How has the significance changed over time? - “While the specific needs caregivers identified as important varied slightly by region, basic needs, such as food (53%), clothing (48%), and school supplies (42%) were the most consistently mentioned...Many families did find assistance, with the most helpful organizations being the Family Resource and Youth Service Centers (FRYSCs) and schools in general (28%).” (https://kyyouth.org/wp-content/uploads/2024/09/Kinship-Across-KY-Report-2024.pdf) - “Nearly 2.7 million young people are being raised by a relative and data suggests kinship rates throughout the Commonwealth are among the highest in the country.” - 2024 (https://uknow.uky.edu/campus-news/meeting-caregivers-where-they-are-kentucky-kinship-resource-center-expanding) - “More than 2.5 million children in the United States were living in kinship care between 2020 and 2022” - 2025 https://journals.sagepub.com/doi/full/10.1177/10443894231204533 - Quarles, R. (2021). Kentucky's Hunger Crisis: A Preventable & Overlooked Tragedy. Ky. J. Equine Agric. & Nat. Resources L., 14, 1, https://uknowledge.uky.edu/cgi/viewcontent.cgi?article=1255&context=kjeanrl. - Background info on hunger crisis in KY - Hawkins, Melissa. Panzera, Anthony. <i>Food insecurity: A key determinant of health</i>. Archives of Psychiatric Nursing, vol. 35, issue 1, 22 October 2020, https://www.psychiatricnursing.org/article/S0883-9417(20)30567-7/fulltext. - Verma, Hema, et. al. <i>Role of Effective Policy and Screening in Managing Pediatric Nutritional Insecurity as the Most Important Social Determinant of Health Influencing Health Outcomes</i>. International Journal of Environmental Research and Public Health, 19 December 2023, https://www.mdpi.com/2072-6643/16/1/5. - Effect of Food Insecurity and Associated Factors on Children - Gianfredi, Vicenza, et. al. <i>Food insecurity is associated with obesity and abdominal obesity among older adults: A cross-sectional analysis of ELSA study</i>. Nutrition, 17 June 2025, https://www.sciencedirect.com.ezproxy.centre.edu/science/article/pii/S0899900725001972. - Outcomes associated with food insecurity among children (obesity) seen later in adulthood - The five largest FNAPs intended to combat hunger and food insecurity are the Supplemental Nutrition Assistance Program (SNAP), the National School Lunch Program (NSLP), the
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	Special Supplemental Nutrition Program for Women, Infants, and Children (WIC), the School Breakfast Program (SBP), and the Summer Food Service Program (SFSP). In 2022, approximately 46.7% of SNAP households, 41.9% of households receiving free or reduced-price school lunches, and 34.6% of WIC households experienced food insecurity https://pmc.ncbi.nlm.nih.gov/articles/PMC10009493/
Public Health Significance: Local or specific to rural communities Findings based on 7-10 peer review research articles (different from above)	- ***We are also currently in communication with Liz Gardner, a member of the FRYSCky Public Policy Committee and the Danville representative for Danville's Grandparents and other Relatives Raising Children support group, and will add that info here once we get it** - Had a meeting scheduled for 12:30 pm on Oct 28th and she canceled, sent our questions via email and waiting on a response - "From 2022, age zero to five reduced by three percent, while representation of age six to 11 and 12 to 18 increased." https://www.chfs.ky.gov/agencies/dcbs/dpp/oohc/Documents/2023%20DCBS%20Relative%20and%20Fictive%20Kin%20Caregiver%20Report.pdf - Dweitt, Emily, et al. <i>Rural SNAP Participants and Food Insecurity: How Can Communities Leverage Resources to Meet the Growing Food Insecurity Status of Rural and Low-Income Residents?</i> International Journal of Environmental Research and Public Health, 19 August 2020, https://www.mdpi.com/1660-4601/17/17/6037. - Discusses rural KY and public health issues regarding food insecurity and related outcomes and the factors that go into determining the differences between rural and urban food insecurity statistics.
Feasible solutions based on peer review research	Hypothesize potential solutions to your public health problem. ● Provide resources for community garden either at the food bank or somewhere nearby that everyone could use and take care of ○ Could also be one near the schools so kids could go there after and snack ○ Also good education opportunity ● Pediatrician/hospital screening for food insecurity ● Provide information on local resources (USDA has a scale with 18 questions) ○ https://publications.aap.org/pediatrics/article/136/5/e1431/33896/Promoting-Food-Security-for-All-

	<u>Children-</u> example of this being carried out What are successful interventions that have worked in other populations that could be implemented in our community? - Community partnership to assist in fresh food supply chains has been beneficial in some communities (https://www.psychiatricnursing.org/article/S0883-9417%2820%2930567-7/fulltext) What are the challenges to implementing some of these potential solutions? - Money is always a barrier to initiatives that involve food insecurity
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GROUP PROJECT OUTLINE: MENTAL HEALTH

PRIMARY OUTCOME	What is the primary outcome? Chronic depression How are you <i>specifically</i> defining this outcome? Two or more years of feeling depressed or sad on most days (37.4% for Lincoln County)
PRIMARY EXSPORE(S)	What is associated with your primary outcome? Lack of leisure time physical activity (which is high stress/obligations and/or access barriers) - The main exposure of having less leisure time physical activity is causing the primary outcome of chronic depression.
Health Disparity	Based on individual characteristics (gender, age, LGBTQ+, SES, income, education, neighborhood, etc.), what is the health disparity? Health Disparity: residents do not have equal access to facilities for physical activity which is causing an increase in chronic depression Kentucky residents with lack of leisure time physical activity due to What is the magnitude and the direction of the health disparity? Individuals with lower SES, who live in neighborhoods with lower overall SES, and who work in high stress jobs *high stress/obligations to family and work
Public Health Significance: Big Picture Findings based on 7-10 peer review research articles	Why do we care? How has the significance changed over time? Exercise is medicine → The WHO reported that around 3.2 million deaths each year are attributable to physical inactivity. → Reduced all-cause mortality and reduced risk of developing conditions such as cardiovascular disease and type 2 diabetes in people, including older adults, who exercise regularly at a moderate level. (1) Patients' views of physical activity as treatment for depression: • providing a source of distraction from negative thoughts, and a sense of purpose. (2)

	• low motivation and a lack of confidence as barriers to undertaking physical activity • https://pmc.ncbi.nlm.nih.gov/articles/PMC3063043/pdf/bjgp61-e149.pdf Socioeconomic Status and Depression over the Life Course • Having a good education leads to less depression over the course of life (8) • https://www.jstor.org/stable/2676303?seq=1 Meeting physical activity recommended levels is associated with higher health-related quality of life • Emphasis on the importance of physical activity in quality of life (4) Those meeting recommended physical activity during leisure time gained 1.1 to 2.0 gains in disease-free years • Health benefits more pronounced among those with pre-existing health risks or disadvantaged backgrounds • Increasing population-wide physical activity → reduction of health disparities (5) Increase in light-intensity leisure time physical activity could decrease elevation in depressive symptoms over time; importance of LTPA in moderating depressive symptoms • Study done on college students, but findings could potentially be applied to our context (6) Relationship Between Quality Of Life And Level Of Physical Activity And Family Well-Being https://tmfv.com.ua/journal/article/view/1918/1546 • Relationships with families improved through having physical activity while their were more probelmes in families without physical activities
Public Health Significance: Local or specific to rural communities Findings based on 7-10 peer review research articles (different from above)	Rural Communities Connection → Disparities in physical activity exist by race, socioeconomic status, and rurality. Many rural areas have limited resources for leisure-time physical activity. ◆ Investments in town center connectivity and racially diverse neighborhoods may aid in reducing inequities in leisure-time physical activity among rural residents. (3) Access to sport facilities and access to public open spaces increases LTPA, in turn decreasing depressive symptoms • Study done in Finland • Good definitions of LTPA settings (indoor sports facility, outdoor sports facility, green public open space, built public open space (7)

Higher average mean LTPA among those with higher SES and safe perceptions of neighborhood environment (association varies when adjusted for demographic characteristics)

- Adjusted models revealed that mean LTPA remained lower among those with perceived unsafe neighborhoods
- Influence mediated by self-rated health
- Self-rated health and perception of neighborhood safety are potential barriers to LTPA (9)

Trends in Meeting Physical Activity Guidelines Among Urban and Rural Dwelling Adults — United States, 2008–2017

https://www.cdc.gov/mmwr/volumes/68/wr/mm6823a1.htm?s_cid=mm6823a1_w

- Since 2008, the prevalence of meeting physical activity guidelines increased from 19.4% to 25.3% among urban residents and from 13.3% to 19.6% among rural residents. Among urban residents, all subgroups reported increases, whereas among rural residents, no increases were reported among Hispanics and adults living in the South.

Rural Populations and Health: Determinants, Disparities, and Solutions

<https://pmc.ncbi.nlm.nih.gov/articles/PMC3696077/pdf/PCD-10-E104.pdf>

- Indeed, compared with their urban counterparts, rural communities have higher rates of preventable conditions such as obesity, diabetes, cancer, and injury, and higher rates of related high-risk health behaviors such as smoking, physical inactivity, poor diet, and limited use of seatbelts (2,3). A population 60 million strong, rural individuals are greatly influenced by geography, and so is their health

Built Environment and Health Behaviors Among African Americans

<https://www.ajpmonline.org/action/showPdf?pii=S0749-3797%2808%2900875-1>

- Built environment affects our daily lives; this article looks specifically at African Americans

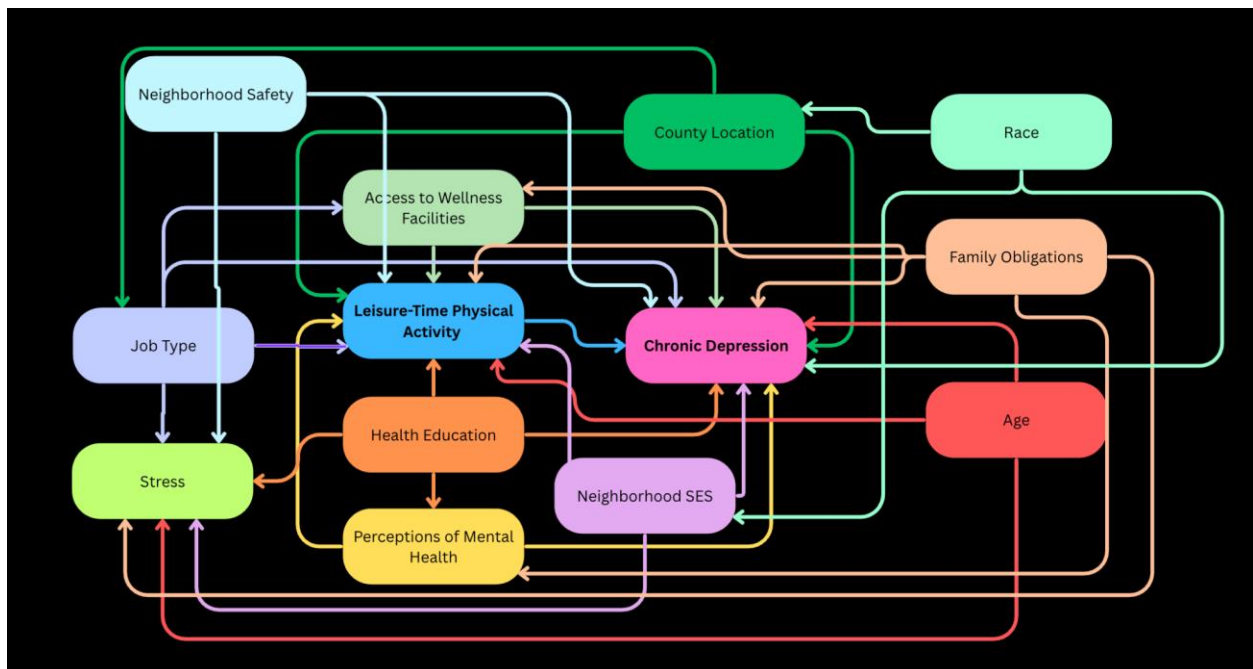
A qualitative examination of home and neighborhood environments for obesity prevention in rural adults

<https://pmc.ncbi.nlm.nih.gov/articles/PMC2631543/pdf/1479-5868-5-65.pdf>

- To date, little research has examined how home and neighborhood environments in rural communities may support or hinder healthy eating and physical activity.

Feasible solutions based on peer review research	Hypothesize potential solutions to your public health problem. - Low/no cost at-home workout plans with minimal equipment - Telehealth connections with coaches or support groups - DREAM: building a wellness center (like the one in Boyle County) in Lincoln County - Working with HS or local churches to start up fitness classes or recreational events What are successful interventions that have worked in other populations that could be implemented in our community? - What are the challenges to implementing some of these potential solutions? - Cost of building facility - Getting buy-in - Staffing
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Sources:

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2. ..

3. Kegler, Michelle C et al. "Inequities in Physical Activity Environments and Leisure-Time Physical Activity in Rural Communities." *Preventing chronic disease* vol. 19 E40. 7 Jul. 2022, doi:10.5888/pcd19.210417
 - a. SEE WHO THEY CITED AND WHO THEY ARE BUILDING OFF OF
 - b. GOOGLE SCHOLAR, WHO HAS CITED THEM
 - c. CLOSELY READ THE STUDY AND SEE WHAT WAS THE POPULATION
 - i. How does this population compare to the EM service area?
4. Vuillemin, A., Boini, S., Bertrais, S., Tessier, S., Oppert, J. M., Hercberg, S., ... & Briançon, S. (2005). Leisure time physical activity and health-related quality of life. *Preventive medicine*, 41(2), 562-569.
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6. Shen, B., Cui, G., & Bo, J. (2025). How does change in leisure-time physical activity influence the growth trajectory of depressive symptoms in college students?. *Journal of American College Health*, 73(3), 1205-1212.
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8. Miech, R. A., & Shanahan, M. J. (2000). Socioeconomic Status and Depression over the Life Course. *Journal of Health and Social Behavior*, 41(2), 162–176. <https://doi.org/10.2307/2676303>
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